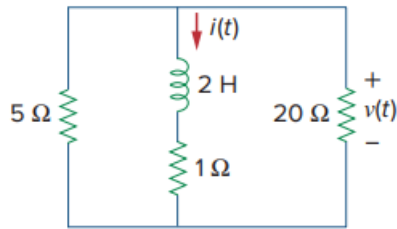


Problem

Find $i(t)$ and $v(t)$ for $t > 0$ in the circuit of Fig. 7.102 if $i(0) = 10$ A.

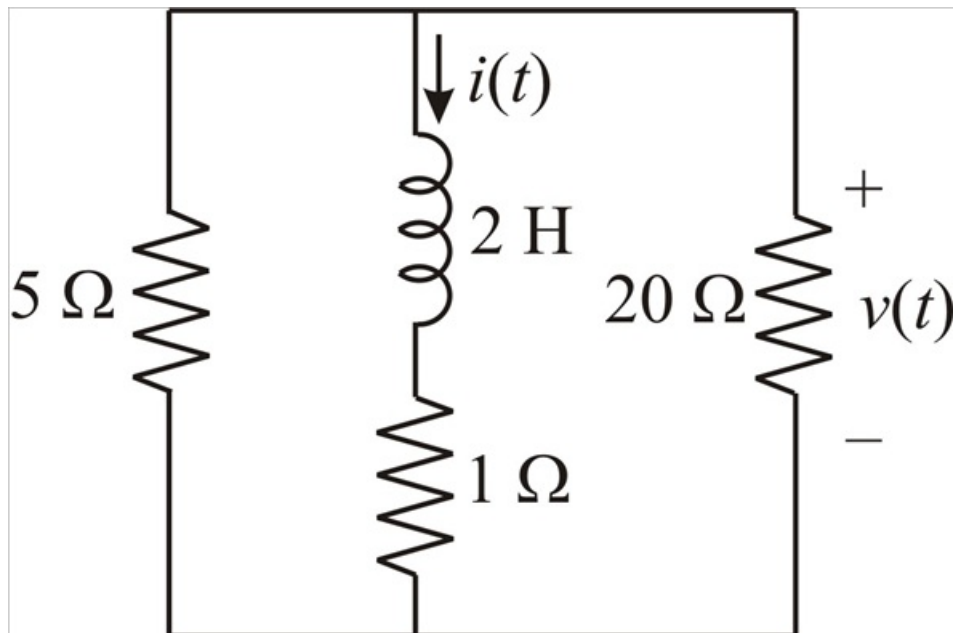
Fig. 7.102:



Step-by-step solution

Step 1 of 10

Given circuit diagram is



Step 2 of 10

Given data:

Initial inductor current is, $i(0) = 10$ A

From the above circuit

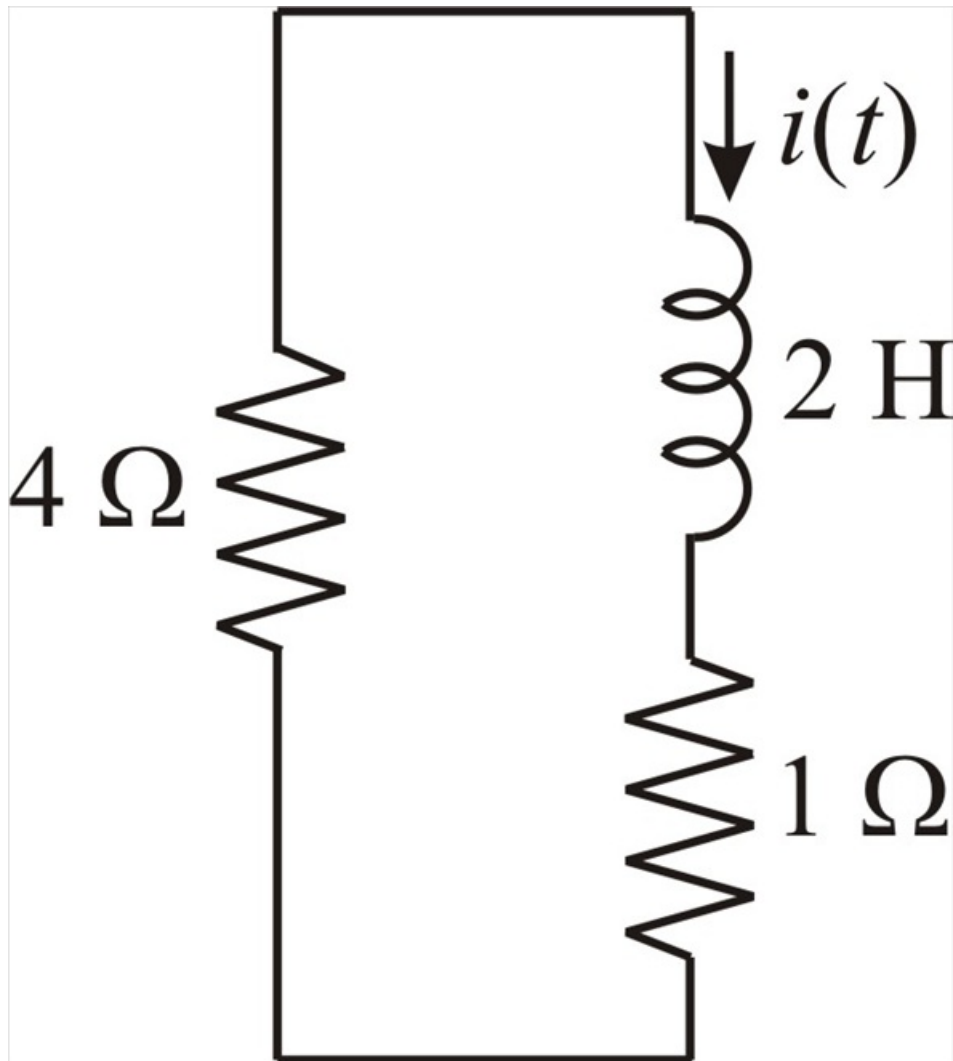
The $5\ \Omega$ resistor and $20\ \Omega$ resistor are in parallel. Its equivalent resistance is

$$\begin{aligned} R_{eq1} &= 20 \parallel 5 \\ &= \frac{(20)(5)}{20 + 5} \\ &= \frac{100}{25} \end{aligned}$$

$$R_{eq1} = 4$$

Hence, we get $R_{eq1} = 4\ \Omega$

Above circuit can be reduced as



From the above circuit

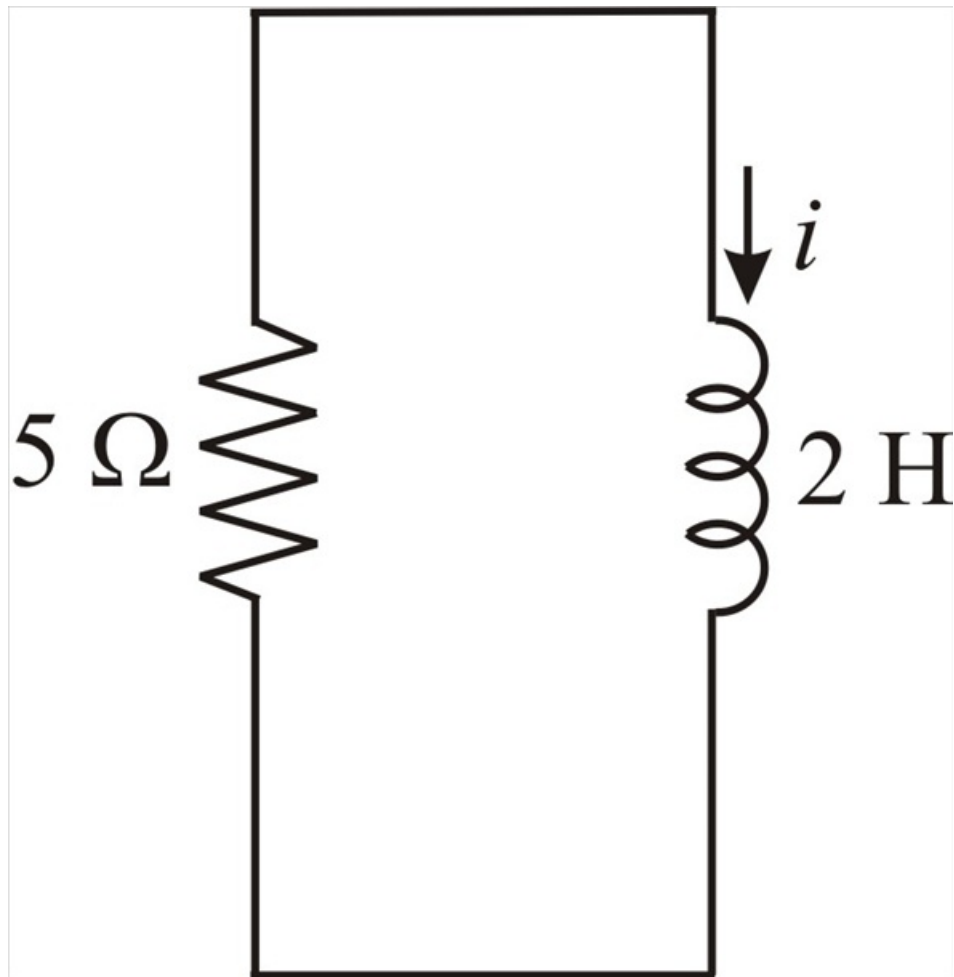
The 4Ω resistor and 1Ω resistor are in series. Its equivalent resistance is

$$R_{eq2} = 4 + 1$$

$$R_{eq2} = 5$$

Hence, we get $R_{eq2} = 5\Omega$.

Above circuit can be reduced as



Step 6 of 10

Applying Kirchhoff's voltage law to the above circuit, we get

$$5i + 2 \frac{di}{dt} = 0$$

$$\frac{di}{dt} + \frac{5}{2}i = 0$$

$$\frac{di}{dt} + 2.5i = 0 \dots\dots (1)$$

Step 7 of 10

Equation-(1) is a first-order differential equation, since only the first derivative of i is involved.

To solve it, we rearrange the terms as

$$\frac{di}{dt} = -2.5i$$

$$\frac{di}{i} = -2.5 dt$$

Integrating both sides to above equation, we get

$$\ln i = -2.5t + \ln A$$

Where $\ln A$ is the integration constant. Thus,

$$\ln i - \ln A = -2.5t$$

$$\ln \frac{i}{A} = -2.5t$$

Taking powers of e produces

$$\frac{i}{A} = e^{-2.5t}$$

$$i = Ae^{-2.5t}$$

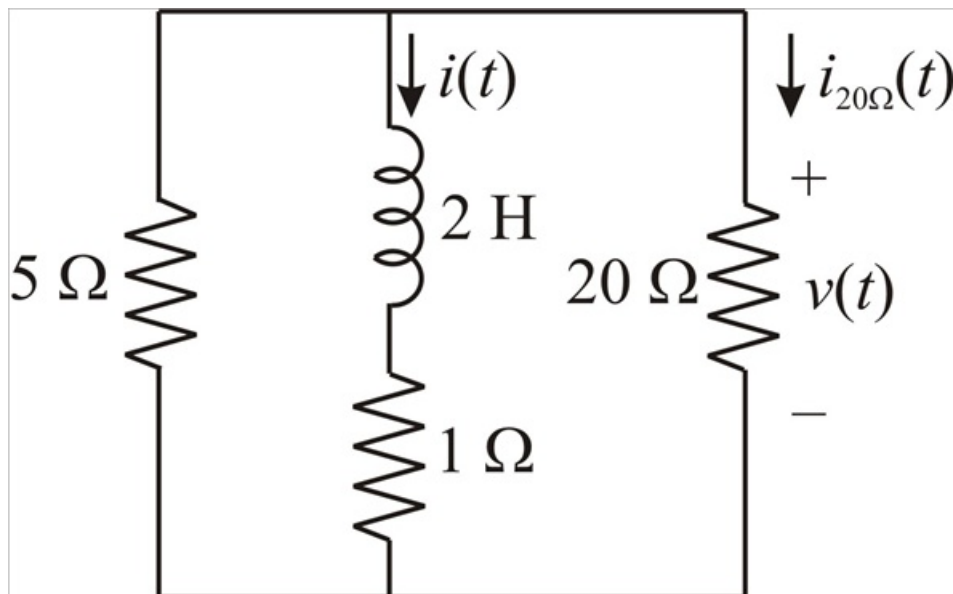
But from the initial conditions, $i(0) = A = I_o = 10 \text{ A}$. Hence,

$$i = 10e^{-2.5t}$$

Therefore, the inductor current is $i = 10e^{-2.5t} u(t) \text{ A}$.

Step 8 of 10

For finding voltage across 20Ω resistor, consider the below circuit



Step 9 of 10

Using current division rule to the above circuit, the current through 20Ω resistor is

$$i_{20\Omega}(t) = -i(t) \left[\frac{5}{5+20} \right]$$

$$= -(10e^{-2.5t}) \left[\frac{5}{25} \right]$$

$$i_{20\Omega}(t) = -2e^{-2.5t}$$

Hence, we get $i_{20\Omega}(t) = -2e^{-2.5t} u(t) \text{ A}$

From the above circuit,

$$\begin{aligned}v(t) &= 20 i_{20\Omega}(t) \\&= 20(-2e^{-2.5t}) \\v(t) &= -40e^{-2.5t}\end{aligned}$$

Therefore, the voltage across 20Ω resistor is $v(t) = -40e^{-2.5t}u(t)\text{V}$.